

RISTRASK V, Sweden

WMO: 022560

Lat: 64.7343N

Lon: 17.4034E

Elev: 575

StdP: 94.61

Time Zone: 1.00 (EUC)

Period: 01-16

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.1	-19.2	-23.7	0.5	-19.7	-21.7	0.6	-17.9	6.5	-4.2	5.8	-5.4	1.7	50	0.703

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.1	23.1	13.9	21.2	13.3	19.1	12.1	16.1	19.8	14.8	17.7	13.6	16.6	2.6	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.6	11.1	17.0	13.5	10.4	15.9	12.4	9.6	14.9	47.1	19.8	43.3	17.7	40.1	16.7	19.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
6.4	5.6	5.1	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			WB	-23.1	16.0	2.4	1.5	-24.8	17.1	-26.2	17.9	-27.5	18.8	-29.3	19.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	1.3	-8.6	-8.3	-4.7	0.4	5.1	10.1	13.1	11.9	6.5	0.6	-3.7	-7.0
	DBStd	8.69	5.25	5.62	4.51	3.62	3.46	4.06	3.14	3.35	3.45	3.95	4.29	5.30
	HDD10.0	3406	577	512	455	288	159	48	6	18	112	292	412	528
	HDD18.3	6211	835	745	713	538	412	250	164	202	354	550	662	786
	CDD10.0	241	0	0	0	0	6	50	103	75	8	0	0	0
	CDD18.3	4	0	0	0	0	0	2	2	1	0	0	0	0
	CDH23.3	34	0	0	0	0	0	16	16	2	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	2.5	1.8	2.2	2.9	2.9	3.0	2.8	2.5	2.3	2.6	2.7	2.1	1.9
Precipitation	PrecAvg	643	46	30	31	31	44	69	98	88	56	55	50	44
	PrecMax	891	93	67	54	58	78	120	165	174	148	140	103	84
	PrecMin	454	5	4	13	8	10	19	33	32	18	21	21	7
	PrecStd	104	18	15	12	15	20	27	34	49	29	26	26	17
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.7	3.7	7.8	13.5	18.6	25.1	25.2	23.6	17.7	10.1	7.2	2.5
		MCWB	1.7	1.8	3.0	5.7	8.8	12.3	16.8	15.0	11.6	6.9	6.4	0.9
	2%	DB	1.2	2.2	4.4	9.6	15.7	23.0	23.1	22.0	15.0	8.3	4.6	1.5
		MCWB	0.6	1.2	1.5	3.6	7.6	11.7	14.7	14.1	10.9	6.9	3.5	0.5
	5%	DB	-0.2	0.3	2.9	7.9	13.1	19.9	21.2	20.0	13.1	7.3	3.1	0.5
		MCWB	-0.7	-0.5	0.6	2.9	6.4	10.5	13.7	13.6	10.4	6.2	2.3	0.0
	10%	DB	-1.7	-0.9	1.6	6.0	10.7	17.2	19.2	17.8	11.6	6.2	1.6	-0.4
		MCWB	-2.1	-1.7	-0.1	2.1	5.1	9.5	12.6	12.4	9.3	5.3	1.1	-0.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.1	2.4	3.4	6.9	10.8	13.6	18.1	17.0	13.5	8.4	6.5	1.7
		MCDB	2.7	3.3	5.4	11.0	15.5	20.5	23.9	21.3	15.4	8.9	7.2	2.0
	2%	WB	0.6	1.2	2.0	4.7	9.0	12.5	16.6	15.8	12.1	7.3	4.0	0.9
		MCDB	1.1	2.1	4.0	8.4	13.2	20.1	20.8	19.5	13.4	8.0	4.3	1.2
	5%	WB	-0.6	-0.3	1.1	3.5	7.4	11.5	15.4	14.6	11.0	6.4	2.5	0.1
		MCDB	-0.3	0.2	2.5	6.9	11.1	18.4	18.6	17.4	12.3	7.1	2.9	0.4
	10%	WB	-2.1	-1.5	0.0	2.4	6.2	10.3	14.1	13.6	10.0	5.4	1.1	-0.8
		MCDB	-1.8	-1.0	1.2	5.3	9.7	16.5	17.0	16.3	11.0	6.1	1.4	-0.4
Mean Daily Temperature Range	5% DB	MDBR	4.8	5.3	6.3	7.0	7.5	8.6	8.1	7.8	6.4	4.5	3.8	4.3
		MCDBR	4.6	5.2	6.8	8.6	10.8	11.8	10.4	10.6	8.4	5.4	4.1	4.7
		MCWBR	4.6	4.6	4.8	4.6	5.7	5.5	4.8	5.1	5.0	4.0	3.9	4.5
	5% WB	MCDBR	4.7	5.0	6.6	7.6	8.7	10.5	9.3	9.1	7.2	5.0	4.1	4.6
		MCWBR	4.7	4.8	5.1	4.7	5.4	5.3	5.6	5.2	5.3	4.1	4.1	4.6
Clear-Sky Solar Irradiance	taub	0.178	0.251	0.280	0.311	0.320	0.320	0.331	0.318	0.293	0.275	0.336	N/A	
	taud	1.956	2.207	2.270	2.343	2.499	2.526	2.502	2.526	2.584	2.458	2.796	N/A	
	Ebn at Noon	379	666	809	865	894	899	875	856	812	652	364	N/A	
	Edh at Noon	23	54	80	95	90	90	90	80	61	43	21	N/A	
All-Sky Solar Radiation	RadAvg	0.13	0.69	2.08	3.80	4.71	5.24	4.79	3.58	2.04	0.83	0.20	0.04	
	RadStd	0.01	0.11	0.21	0.30	0.54	0.40	0.54	0.38	0.25	0.13	0.04	0.00	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(47)	Regional (42 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page